NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Estimation Techniques cont…



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CONCEPTS COVERED

- Parametric models
  - Some more examples on Albrecht/IFPUG function points
  - Symons/Mark II function points
  - COSMIC function points



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Parametric models for Size

1. Albrecht/IFPUG function points

2. Symons/Mark II function points

3. COSMIC function points

4. COCOMO 81 and COCOMO II





Example on Albrecht/IFPUG FP

Consider a project with the following functional units:

- 50 user inputs, 40user outputs, 35 user enquiries, 6 user files, 4 external interfaces. Assume all complexity adjustment factors and weighting factors are average.
- Compute the function points for the project.
- Suppose that program needs 70 LOC per FP.
- Find out the size of complete project.





Solution

- UFP= 50 * 4 + 40 * 5 + 35 * 4 + 6 * 10 + 4 * 7
  = 200+200+140+60+28 = 628
- VAF=0.65 + 0.01 (14 * 3) = 1.07
- AFP= UFP * VAF=628 * 1.07 = 672
- Size = FP * (LOC per FP) = 672 * 70 = 47040 LOC

Another example

- Compute the function-point value for a project with the following information-domain characteristics.
  - Number of user Inputs: 32
  - Number of User Output: 60
  - Number of User Inquiries: 24
  - Number of Files: 8
  - Number of External interface:2
- Assume that all complexity adjustment values are average.





Solution

- UFP = 32x4 + 60x5 + 24x4 + 8x10 + 2x7 = 128 + 300 + 96 + 80 + 14
  = 618
- VAF = 0.65 + 0.01x(14x3) = 0.65+0.42 = 1.07
- AFP = UFP x VAF = 618x1.07 = 661.26

Function points Mark II

- Developed by Charles R. Symons
- ‘Software sizing and estimating - Mk II FPA’, Wiley & Sons, 1991.
- Builds on work by Albrecht
- Developed in parallel to IFPUG FPs
- A simpler and more usable method





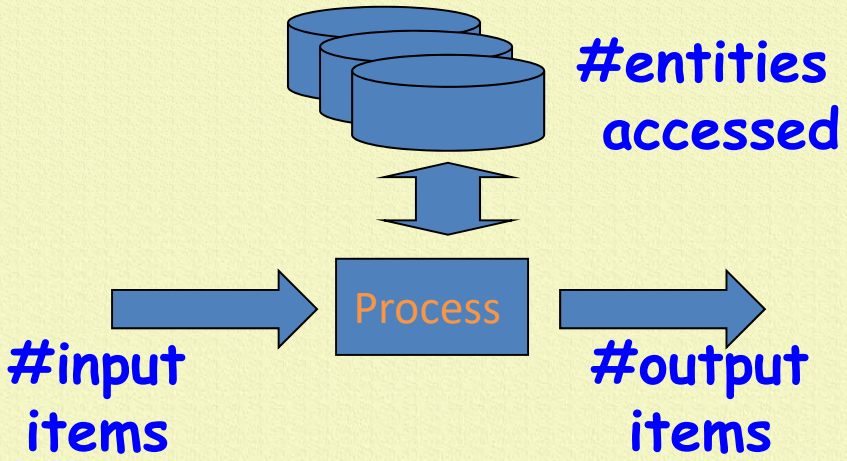
Function points Mk II cont…

- For each transaction, count
  - data items input (N_i)
  - data items output ($N_{o)}$
  - entity types accessed (N_e)





Function points Mk II cont…



- Simpler than FP
- Widely used in UK

FP count = N_i * 0.58 + N_e * 1.66 + N_o * 0.26

Function points for embedded systems

- Mark II and IFPUG function points were designed for information systems, and not suitable for embedded systems:
  - Dominated by input and output operations
- COSMIC Full FPs attempt to extend the concept to embedded systems
- In an embedded system, its features will be hidden, because the software's user will probably not be a human,
  - it will be hardware device or another software component



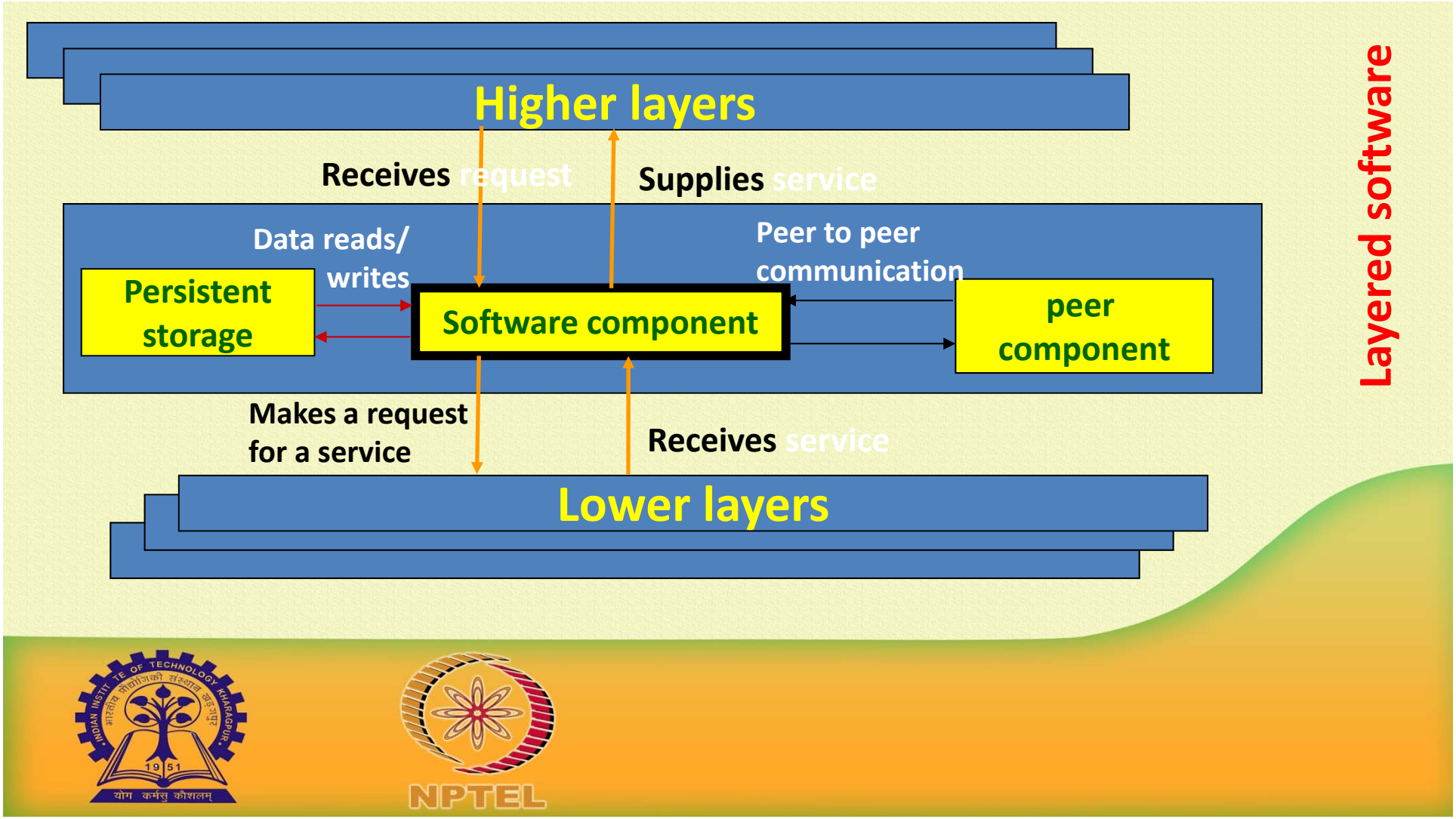


Function points for embedded systems cont...

- COSMICS deals with this by decomposing the system architecture into a hierarchy of software layers.
- Embedded software is seen as being in a particular 'layer' in the system
  - Communicates with other layers and also other components at same level
- The software component to be sized, can receive requests for services from layers above and can request services from those below it.



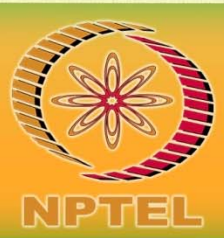



Higher layers
Receives request
Supplies service
Data reads/
writes
Persistent
storage
Software component
Peer to peer
communication
peer
component
Makes a request
for a service
Receives service
Lower layers
Layered software

Function points for embedded systems cont...

- This identifies the boundary of the software component to be accessed and thus the points at which it receives inputs and transmits outputs.
- Inputs and outputs are aggregated into data groups, where each group brings together data items that relate to the same object of interest.





COSMIC FPs

Data groups can move about in 4 ways as follows.

- Entries: movement of data into software component from a higher layer or a peer component
- Exits: movements of data out
- Reads: data movement from persistent storage
- Writes: data movement to persistent storage

Each counts 1 ‘COSMIC functional size unit’ (Cfsu).





COSMIC FPs cont ...

- The overall FFP count is derived by simply adding up the counts for each of the 4 types of data movement.
- COSMIC FFPs have been incorporated into ISO standard.





COSMIC FPs cont ...

Cons:

- Does not take into account any processing of the data groups once they have been moved into the software component.
- Not recommended for use in systems involving complex mathematical algorithms.

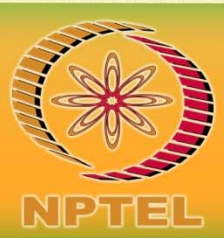




Function Points: Pros and Cons

- Pros:
  - Language independent
  - Understandable by client
  - Simple modeling
  - Hard to fudge
  - Visible feature creep
- Cons:
  - Labor intensive
  - Extensive training
  - Inexperience may result in inaccuracy
  - Weighted to file manipulation and transactions
  - Errors may be introduced by single person, multiple raters advised…
  - Does not consider algorithmic complexity of a function.





Feature Point Metric

- FP metric Suffers from a major drawback:
  - the size of a function is considered to be independent of its complexity.
- In order to overcome this problem, an extension to the function point metric, called feature point metric has been proposed.
  - considers an extra parameter:
    - Algorithm Complexity.

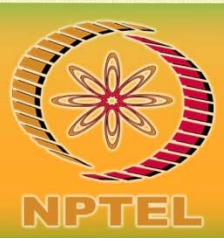




Feature Point Metrics cont ...

- This parameter ensures that
  - the computed size using the feature point metric reflects the fact that
    - higher the complexity of a function, the greater the effort required to develop it
- Therefore, it should have larger size compared to a simpler function.

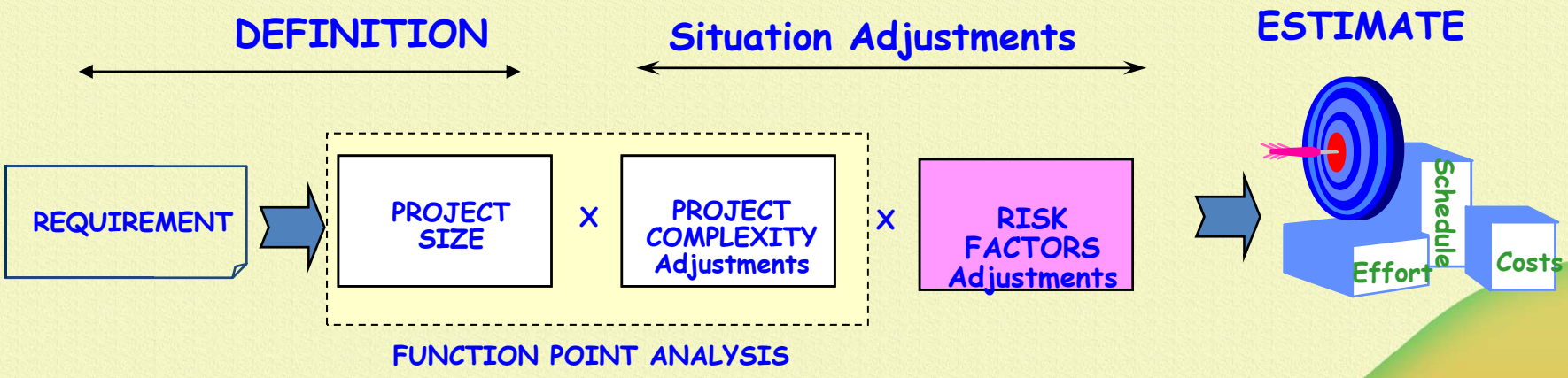




FP/SLOC Conversion

Language	Median SLOC/function point
C	104
C++	53
HTML	42
JAVA	59
Perl	60
J2EE	50
Visual Basic	42

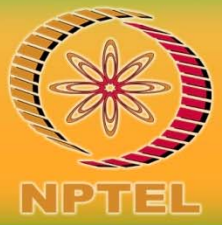
Accurate Size Estimation



Object Points

- Object points has nothing to do with object-oriented programming
- Number of object points is estimated based on
  - Number of separate screens displayed
  - Number of reports that are produced
  - Number of modules in the code
- Object points are simpler to estimate & take GUI into account





Summary

- Solved some more examples on Albrecht/IFPUG function points
- Discussed Symons/Mark II function points and COSMIC function points.
- Also, discussed the concepts feature points and object points.





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






Thank you
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